

LAMIVOX 10mg/ml (Lamivudine Oral Solution 10mg/ml)

Description: A clear, colorless to pale yellow, strawberry-banana flavored liquid. Each ml contains Lamivox 10mg.

Pharmacodynamics:

Lamivudine is a potent, selective inhibitor of HIV-1 and HIV-2 replication in vitro. It is also active against zidovudine-resistant clinical isolates of HIV. Lamivudine is metabolized intracellularly to the 5'-triphosphate, the active moiety, which has an intra-cellular half-life of 16 to 19 h. lamivudine 5'-triphosphate is a weak inhibitor of the RNA and DNA dependent activities of HIV reverse transcriptase, its main mode of action is as a chain terminator of HIV reverse transcription. Lamivudine has been shown to act additively or synergistically with other anti-HIV agents, particularly zidovudine, inhibiting the replication of HIV in cell culture.

Lamivudine does not interfere with cellular deoxynucleotide metabolism and has little effect on mammalian cell and mitochondrial DNA content.

In vitro, lamivudine demonstrates low cytotoxicity to peripheral blood lymphocytes, to established lymphocyte and monocyte-macrophage cell lines, and to a variety of bone marrow progenitor cells in vitro. Lamivudine therefore has, in vitro, a high therapeutic index.

HIV-1 resistance to lamivudine involves the development of a M184V amino acid change close to the active site of the viral reverse transcriptase (RT). This variant arises both in vitro and in HIV-1 infected patients treated with lamivudine-containing antiretroviral therapy. M184V mutants display greatly reduced susceptibility to lamivudine and show diminished viral replicative capacity in vitro. In vitro studies indicate that zidovudine-resistant virus isolates can become zidovudine sensitive when they simultaneously acquire resistance to lamivudine. The clinical relevance of such findings remains, however, not well defined.

Cross-resistance conferred by the M184V RT is limited within the nucleoside inhibitor class of antiretroviral agents. Zidovudine and stavudine maintain their antiretroviral activities against lamivudine-resistant HIV-1. Abacavir maintains its antiretroviral activities against lamivudine-resistant HIV-1 harbouring only the M184V mutation. The M184V RT mutant shows less than 4-fold decrease in susceptibility to didanosine and zalcitabine; the clinical significance of these findings is unknown. In vitro susceptibility testing has not been standardized and results may vary according to methodological factors.

In clinical trials, lamivudine in combination with zidovudine has been shown to reduce HIV-1 viral load and to increase CD4 cell count. Clinical end-point data indicate that lamivudine in combination with zidovudine alone or in combination with zidovudine containing treatment regimens results in a significant reduction in the risk of disease progression and mortality.



Reduced in vitro sensitivity to lamivudine has been reported for HIV isolates from patients who have received Lamivudine.

Evidence from clinical studies show that lamivudine plus zidovudine delays the emergence of zidovudine-resistant isolates in individuals with no prior anti-retroviral therapy.

Lamivudine has been widely used as a component of antiretroviral combination therapy with other antiretroviral agents of the same class (nucleoside reverse transcriptase inhibitors) or different classes (protease inhibitors, non-nucleoside reverse transcriptase inhibitors).

Multiple drug antiretroviral therapy containing lamivudine has been shown to be effective in antiretrovirally-naïve patients as well as in patients presenting with viruses containing the M184V mutations. The relationship between in vitro susceptibility of HIV to lamivudine and the clinical response to therapy remain under investigation.

Post-exposure prophylaxis (PEP):

Internationally recognized guidelines (Centre for Disease Control and Prevention – June 1998), recommend that in the event of accidental exposure to HIV infected blood e.g. from a needle stick injury, a combination of zidovudine and lamivudine should be administered promptly (within 1 to 2 h). In cases of higher risk of the infection a protease inhibitor should be included in the regimen. It is recommended that antiretroviral prophylaxis be continued for four weeks. No controlled clinical studies have been carried out in post-exposure prophylaxis and supporting data limited. Seroconversion may still occur despite prompt treatment with antiretroviral agents.

Pharmacokinetic:

Absorption: Lamivudine is well absorbed from the gastrointestinal tract, and the bioavailability of oral lamivudine in adults is normally between 80 and 85 %. Following oral administration, the mean time ($t_{\max}$) to maximal serum concentrations ($C_{\max}$) is about an hour. At therapeutic dose levels i.e. 4mg/kg/day (as two 12-hourly doses), $C_{\max}$ is in the order of 1 to 1.9 micrograms/ml.

Co-administration of lamivudine with food resulted in a delay of $t_{\max}$ and a lower $C_{\max}$ (decreased by 47 %). However, the extent (based on the AUC) of lamivudine absorbed was not influenced. No dose adjustment is needed when co-administered with food.

Metabolism and Elimination: Lamivudine mean systemic clearance is approximately 0.32 l/h/kg, with predominantly renal clearance (greater than 70%) via the organic cationic transport system, and little (less than 10%) hepatic metabolism.

The active moiety, intracellular lamivudine triphosphate, has a prolonged terminal half-life in the cell (16 to 19 hours) compared to the plasma lamivudine half-life (5 to 7 hours). The likelihood of adverse interactions between lamivudine with other medicinal products is low due to limited metabolism and plasma protein binding and almost complete renal elimination of unchanged lamivudine.



Indication:

Lamivox is indicated as part of antiretroviral combination therapy for the treatment of Human Immunodeficiency Virus (HIV) infected adults and children.

Recommended Dose:

Lamivudine therapy should be initiated by a physician experienced in the management of HIV infection.

Adults and adolescents over 12 years of age: The recommended dose of Lamivudine is 300 mg daily. This may be administered as 150 mg (15ml) twice daily or 300 mg (30ml) once daily.

Children (Three months to 12 years of age): The recommended dose is 4 mg/kg twice daily up to a maximum of 300 mg daily.

Less than three months of age: The limited data available are insufficient to propose specific dosage recommendations.

Elderly: No specific data are available; however, special care is advised in this age group due to age associated changes such as the decrease in renal function and alteration of haematological parameters.

Renal impairment: Lamivudine plasma concentrations (AUC) are increased in patients with moderate to severe renal impairment due to decreased clearance. The dosage should therefore be reduced for patients with a creatinine clearance of less than 50 ml/min as shown in the table below. The same percentage reduction in dose applies for pediatric patients with renal impairment.

Dosing Recommendations – Adults and adolescents over 12 years:

Creatinine Clearance

(ml/min)	First Dose	Maintenance Dose
30 to less than 50	150 mg [#]	150 mg [#] Once daily
15 to less than 30	150 mg [#]	100 mg (10ml) Once daily
5 to less than 15	150 mg [#]	50 mg (5ml) Once daily
Less than 5	50 mg (5ml)	25 mg (2.5ml) Once daily

[#] 150 mg: 15 ml oral solution



Dosing Recommendations – Children from 3 months to 12 years:

Creatinine Clearance (ml/min)	First Dose	Maintenance Dose
30 to less than 50	4 mg/kg	4 mg/kg once daily
15 to less than 30	4 mg/kg	2.6 mg/kg once daily
5 to less than 15	4 mg/kg	1.3 mg/kg once daily
Less than 5	1.3 mg/kg	0.7 mg/kg once daily

Hepatic Impairment: No dose adjustment is necessary in patients with moderate or severe hepatic impairment unless accompanied by renal impairment.

Route of administration:

Oral

Contraindications:

Hypersensitivity to lamivudine or to any of the excipients.

Warnings and Precautions:

Lamivudine is not recommended for use as monotherapy.

Patients should be advised that current antiretroviral therapy, including Lamivudine, has not been proven to prevent the risk of transmission of HIV to others through sexual contact or blood contamination. Appropriate precautions should continue to be employed.

Patients receiving Lamivudine or any antiretroviral therapy may continue to develop opportunistic infections and other complications of HIV infection, and there they should remain under close clinical observation by physicians experienced in the treatment of patients with associated HIV diseases.

Renal impairment

Lamivudine plasma concentrations (AUC) are increased in patients with moderate to severe renal impairment due to decreased clearance. The dose should therefore be adjusted (see Recommended Dose).

Pancreatitis

Pancreatitis has been observed in some patients receiving Lamivudine. However, it is unclear whether this was due to treatment with the medicinal product or to the underlying HIV disease. Pancreatitis must be considered whenever a patient develops abdominal pain, nausea, vomiting or elevated biochemical markers. Discontinue use of Lamivudine until diagnosis of pancreatitis is excluded.



Lactic acidosis/severe hepatomegaly with steatosis

Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of antiretroviral nucleoside analogues either alone or in combination, including lamivudine. A majority of these cases have been in women.

Clinical features which may be indicative of the development of the lactic acidosis include generalized weakness, anorexia, and sudden unexplained weight loss, gastrointestinal symptoms and respiratory symptoms (dyspnoea and tachypnoea).

Caution should be exercised when administering Lamivudine to any patient, and particularly to those with known risk factors for liver disease. Treatment with Lamivudine should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis or hepatotoxicity (which may include hepatomegaly and steatosis even in the absence of marked transaminase elevations).

Fat redistribution

Redistribution/accumulation of body fat, including central obesity, dorsocervical fat enlargement (buffalo hump), peripheral wasting, facial wasting, breast enlargement, elevated serum lipid and blood glucose levels have been observed either separately or together in some patients receiving combination antiretroviral therapy.

Whilst all members of the PI and NRTI classes of medicinal products have been associated with one or more of these specific adverse events, linked to a general syndrome commonly referred to as lipodystrophy, data indicate that there are differences in the risk between individual members of the respective therapeutic classes.

In addition, the lipodystrophy syndrome has a multi-factorial aetiology; with for example HIV disease status, older age and duration of antiretroviral treatment all playing important, possibly synergistic roles.

The long-term consequences of these events are currently unknown.

Clinical examination should include evaluation for physical signs of fat redistribution. Consideration should be given to the measurement of serum lipids and blood glucose. Lipid disorders should be managed as clinically appropriate.

Patients co-infected with Hepatitis B virus

Clinical trial and marketed use of Lamivudine, have shown that some patients with chronic hepatitis B virus (HBV) disease may experience clinical or laboratory evidence of recurrent hepatitis upon discontinuation of Lamivudine, which may have more severe consequences in patients with decompensated liver disease. If Lamivudine is discontinued in a patient with HIV and HBV coinfection, periodic monitoring of both liver function tests and markers of HBV replication should be considered.



Interactions with Other Medicaments

The likelihood of metabolic interactions is low due to limited metabolism and plasma protein binding and almost complete renal elimination of unchanged lamivudine.

Lamivudine is predominantly eliminated by active organic cationic secretion. The possibility of interactions with other medicinal products administered concurrently should be considered, particularly when their main route of elimination is active renal secretion via the organic cationic transport system e.g. trimethoprim. Other active substances (e.g. ranitidine, cimetidine) are eliminated only in part by this mechanism and were shown not to interact with lamivudine.

Active substances shown to be predominantly excreted either via the active organic anionic pathway, or by glomerular filtration are unlikely to yield clinically significant interactions with lamivudine.

Zidovudine: A modest increase in C_{max}(28%) was observed for zidovudine when administered with lamivudine, however overall exposure (AUC) was not significantly altered. Zidovudine had no effect on the pharmacokinetics of lamivudine.

Trimethoprim/sulphamethoxazole: Administration of trimethoprim/sulphamethoxazole 160mg/800mg (co-trimoxazole) causes a 40% increase in lamivudine exposure because of the trimethoprim component. However, unless the patient has renal impairment, no dosage adjustment of lamivudine is necessary. Lamivudine has no effects on the pharmacokinetics of trimethoprim or sulphamethoxazole. The effect of co-administration of lamivudine with higher doses of co-trimoxazole for the treatment of *Pneumocystis jirovecii* (P. carinii) pneumonia and toxoplasmosis has not been studied.

Zalcitabine: Lamivudine may inhibit the intracellular phosphorylation of zalcitabine when the two medicinal products are used concurrently. Lamivudine is therefore not recommended to be used in combination with zalcitabine

Pregnancy and Lactation:

There are limited data available on the safety of Lamivudine in human pregnancy. Studies in humans have confirmed that lamivudine crosses the placenta. Use in pregnancy should be considered only if the benefit outweighs the risk. Although the results of animal studies are not always predictive of human response, the findings in the rabbit suggest a potential risk of early embryonic loss.

There have been reports of mild, transient elevations in serum lactate levels, which may be due to mitochondrial dysfunction, in neonates and infants exposed in utero or peri-partum to nucleoside reverse transcriptase inhibitors (NRTIs). The clinical relevance of transient elevations in serum lactate is unknown. There have also been very rare reports of developmental delay, seizures and other neurological disease. However, a causal relationship between these events and NRTI exposure in utero or peri-partum has not been established. These findings do not affect current recommendations to use antiretroviral therapy in pregnant women to prevent vertical transmission of HIV. Health experts recommend that where possible women infected with HIV do not breast feed their infants in order to avoid the transmission of HIV. Following oral administration



lamivudine was excreted in human breast milk at similar concentrations to those found in serum (1 to 8 micrograms/ml). Since lamivudine and the virus pass into breast milk it is recommended that mothers taking Lamivudine do not breast feed their infants.

Side effects

The following events have been reported during therapy for HIV disease with disease with Lamivudine alone and in combination with other anti-retroviral agents. With many it is unclear whether they are related to the medicinal products or are as a result of the underlying disease process.

The following convention has been utilized for the classification of undesirable effects: very common, common, uncommon, rare, and very rare.

Blood and lymphatic system disorders

Uncommon : Neutropenia, anaemia, thrombocytopenia.
Very rare : Pure red cell aplasia

Metabolism and nutrition disorders

Common : Hyperlactataemia.
Rare : Lactic acidosis
Redistribution/accumulation of body fat. The incidence of this event is dependent on multiple factors including the particular antiretroviral drug combination.

Nervous system disorders

Common : Headache.
Very rare : Paraesthesia. Peripheral neuropathy has been reported although a causal relationship to treatment is uncertain.

Gastrointestinal disorders

Common : Nausea, vomiting, upper abdominal pain, diarrhoea.
Rare : Pancreatitis, although a causal relationship to treatment is uncertain. Rises in serum amylase.

Hepatobiliary disorders

Uncommon : Transient rises in liver enzymes (AST, ALT).

Skin and subcutaneous tissue disorders

Common : Rash, alopecia.

Musculoskeletal and connective tissue disorders

Common : Arthralgia, muscle disorders.
Rare : Rhabdomyolysis.

General disorders and administration site conditions

Common : Fatigue, malaise, fever.



Symptoms and Treatment of Overdose:

Limited data are available on the consequences of ingestion of acute overdoses in humans. No fatalities occurred, and the patients recovered. No specific signs or symptoms have been identified following such overdose.

If over dosage occurs the patient should be monitored, and standard supportive treatment applied as required. Since lamivudine is dialysable, continuous hemodialysis could be used in the treatment of over dosage, although this has not been studied.

Storage Condition

Do not store above 30°C.

Shelf Life:

18 months

Presentation: Lamivox 10mg/ml (Lamivudine Oral Solution 10mg/ml) is filled in HDPE container 300 ml and packed together with Package insert.

Manufactured by:

Aurobindo Pharma Limited
Unit III, Survey No. 313 & 314,
Bachupally, Bachupally Mandal,
Medchal-Malkajgiri District,
Telangana State, INDIA.

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